Vector Spaces

Math Foundations — Concept 05

1 Motivation

A vector space is the algebraic structure that captures "things you can scale and add together." The motivating examples are arrows in $\mathbb{R}^n$, but the abstraction subsumes polynomial spaces $\mathbb{R}[x]$, function spaces $C([0,1])$ and $L^2(\mathbb{R})$, sequence spaces ℓ^2, and the embedding spaces $\mathbb{R}^d$ that store token representations inside transformers. Linear algebra is the study of these spaces and the structure-preserving maps between them.

2 Axioms

Definition 1 (Vector space)**.** Let $\mathbb{F}$ be a field (e.g. $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{F}_p$). A *vector space over* $\mathbb{F}$ is a set V equipped with operations

$$+ : V \times V \to V, \qquad \cdot : \mathbb{F} \times V \to V$$

such that $(V, +)$ is an abelian group and for all $\alpha, \beta \in \mathbb{F}$ and $u, v \in V$:

1. $\alpha(u + v) = \alpha u + \alpha v$,
2. $(\alpha + \beta)v = \alpha v + \beta v$,
3. $(\alpha\beta)v = \alpha(\beta v)$,
4. $1 \cdot v = v$.

Elements of V are called *vectors*; elements of $\mathbb{F}$ are *scalars*.

Definition 2 (Linear combination, span)**.** Given $v_1, \ldots, v_k \in V$ and $\alpha_1, \ldots, \alpha_k \in \mathbb{F}$, the vector $\sum_i \alpha_i v_i$ is a *linear combination*. The *span* of a set $S \subseteq V$, denoted $\mathrm{span}(S)$, is the set of all finite linear combinations of elements of S; it is the smallest subspace containing S.

Definition 3 (Linear independence)**.** A set $S \subseteq V$ is *linearly independent* if for every finite subset $\{v_1, \ldots, v_k\} \subseteq S$, the equation $\sum_i \alpha_i v_i = 0$ implies $\alpha_1 = \cdots = \alpha_k = 0$. Otherwise S is *linearly dependent*.

Definition 4 (Basis)**.** A *basis* of V is a linearly independent set $B \subseteq V$ with $\mathrm{span}(B) = V$. Equivalently, every $v \in V$ admits a *unique* expression as a finite linear combination of elements of B.

Definition 5 (Dimension)**.** If V admits a finite basis, the *dimension* $\dim_{\mathbb{F}} V$ is the cardinality of any such basis (well-defined by Theorem **??** below). If no finite basis exists, V is *infinite-dimensional*.

3 The Dimension Theorem

Lemma 6 (Steinitz exchange)**.** *Let V be a vector space, let $\{v_1, \ldots, v_m\}$ be a spanning set, and let $\{w_1, \ldots, w_n\}$ be linearly independent. Then $n \le m$, and after relabeling we may replace n of the v_i by the w_j to obtain another spanning set $\{w_1, \ldots, w_n, v_{n+1}, \ldots, v_m\}$.*

Proof. Induct on n. The base case $n = 0$ is trivial. Suppose the result holds for $n-1$: after relabeling, $\{w_1, \ldots, w_{n-1}, v_n, \ldots, v_m\}$ spans V. Since this set spans, write

$$w_n = \sum_{i=1}^{n-1} \alpha_i w_i + \sum_{j=n}^{m} \beta_j v_j.$$

Not all β_j are zero, else $w_n \in \operatorname{span}(w_1, \ldots, w_{n-1})$, contradicting independence of $\{w_1, \ldots, w_n\}$. In particular $m \ge n$. After relabeling, $\beta_n \ne 0$, so

$$v_n = \beta_n^{-1}\Big(w_n - \sum_{i<n} \alpha_i w_i - \sum_{j>n} \beta_j v_j\Big) \in \operatorname{span}(w_1, \ldots, w_n, v_{n+1}, \ldots, v_m).$$

Therefore $\{w_1, \ldots, w_n, v_{n+1}, \ldots, v_m\}$ spans V, completing the induction. □

Theorem 7 (Dimension is well-defined)**.** *Let V be a vector space and suppose V has a finite basis. Then any two bases of V have the same cardinality.*

Proof. Let $B_1 = \{v_1, \ldots, v_m\}$ and $B_2 = \{w_1, \ldots, w_n\}$ be two bases of V. Since B_1 spans V and B_2 is linearly independent, Lemma **??** gives $n \le m$. By symmetry (swap the roles: B_2 spans, B_1 is independent), $m \le n$. Therefore $m = n$. □

Proposition 8 (Existence of a basis, finite case)**.** *If V is spanned by finitely many vectors $v_1, \ldots, v_m$, then some subset of $\{v_1, \ldots, v_m\}$ is a basis of V.*

Proof sketch. If $\{v_1, \ldots, v_m\}$ is linearly independent, done. Otherwise some v_k is a linear combination of the others; remove it. The remaining set still spans. Iterate; the process terminates because the set is finite, yielding a linearly independent spanning set. □

4 Examples

Example 9 ($\mathbb{R}^3$)**.** The standard basis $\{e_1, e_2, e_3\}$ with e_i the i-th coordinate vector gives $\dim \mathbb{R}^3 = 3$. The set $\{(1,1,0), (0,1,1), (1,0,1)\}$ is another basis (the 3×3 matrix with these columns has determinant -2).

Example 10 (Polynomial space)**.** $\mathbb{R}[x]_{\le n} = \{a_0 + a_1 x + \cdots + a_n x^n : a_i \in \mathbb{R}\}$ has basis $\{1, x, x^2, \ldots, x^n\}$, so $\dim = n + 1$.

Example 11 (Function space)**.** $C([0,1])$, the continuous real functions on $[0,1]$, is infinite-dimensional: $\{1, x, x^2, x^3, \ldots\}$ is linearly independent.

5 Connection to ML

In transformer-style models, every token is mapped to a vector in $\mathbb{R}^d$ ($d \approx 4096$ for large models). Attention computes weighted sums (linear combinations) of value vectors; layer norm, residual streams, and gradient updates all operate inside this vector space. The ELF paper (arxiv:2605.10938) defines flow-matching trajectories $x_t = (1-t)x_0 + tx_1$ in $\mathbb{R}^d$, a literal application of the vector-space axioms.